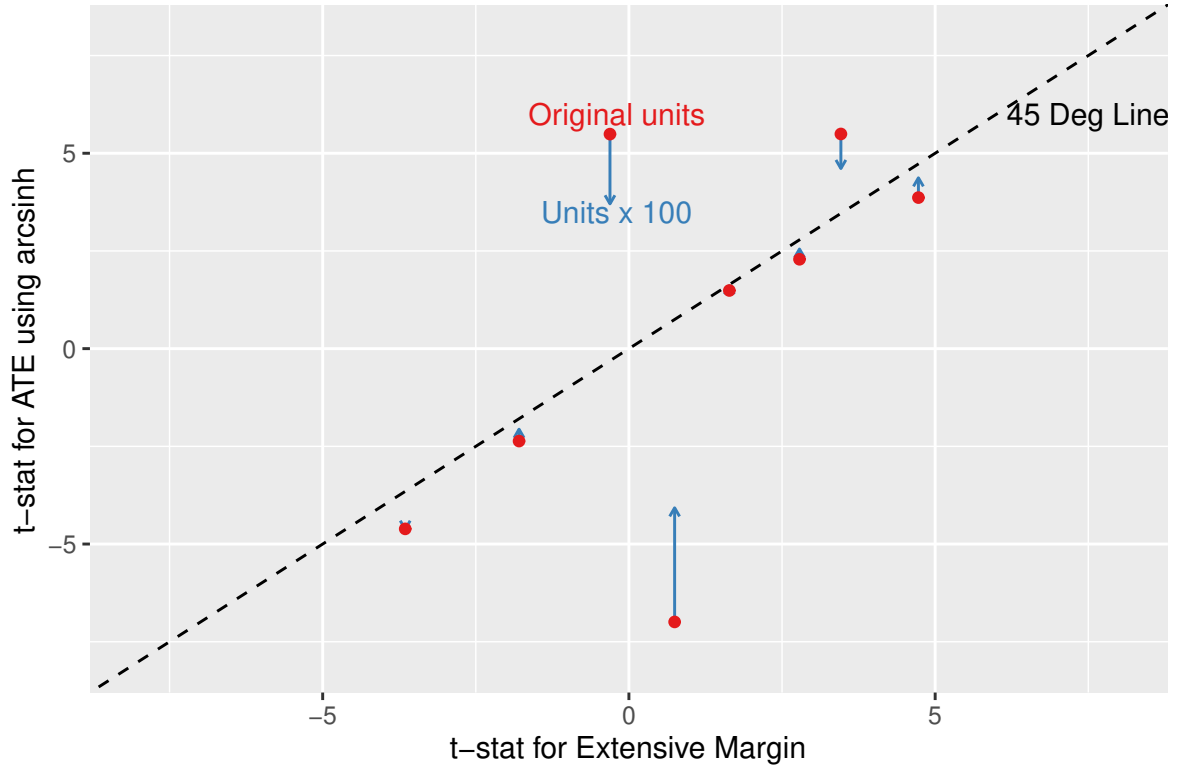


Paper	Interprets Units as %	Original Units	Quote About Percents / Notes
Azoulay et al (2019)	Yes	Publications (yearly)	“In this case, coefficient estimates can be interpreted as elasticities, as an approximation.”
Beerli et al (2021)	Yes	Patent applications (yearly)	“The estimates thus reflect an approximate percentage increase.”
Berkouwer and Dean (2022)	Yes	Weekly expenditure (dollars)	“A 0.50 IHS reduction corresponds to a 39 percent reduction relative to the control group.”
Cabral et al (2022)	Yes	Costs (dollar) per \$10K risk-adjusted covered payroll	Refers to estimates as “the elasticities reported in panel A”
Carranza et al (2022)	Yes	Hours worked (weekly)	“Weekly earnings increase by 34% (Table 1, column 3)”
Faber & Gauber (2019)	Yes	Municipality GDP (1000s of Pesos)	“A one standard deviation increase in tourism attractiveness increases local manufacturing GDP by about 40 percent.”
Hjort and Poulsen (2019)	Yes	KB per second	“We find that cable arrival increases measured speed in connected locations, relative to unconnected locations, by around 35 percent”
Johnson (2020)	Yes	Violations (monthly)	“[T]he regression coefficient estimates the ITT effect of a press release on the percent change in the number of violations. The point estimate (-0.18) is identical to the baseline estimate in percent terms $-0.40/2.29 = 17.5\%$.”
Mirenda et al (2022)	Yes	Contract size (euros)	“The amount of public funds awarded raises by 3.4 percent.”
Norris et al (2021)	Yes	Criminal charges	“We measure both the extensive margin (using a binary indicator for the outcome ever occurring) and the intensive margin (taking the inverse hyperbolic sine, IHS, of the number of times the outcome occurred, so the coefficient is interpreted as a percent change)”
Ager et al (2021)	No interpretation	Wealth (1870 dollars)	
Arora et al (2021)	No interpretation	Publications (yearly)	
Bastos et al (2018)	No interpretation	Sales (yearly, euros)	
Fetzer et al (2021)	No interpretation	Incidents (quarterly)	
Moretti (2021)	No interpretation	Patents (yearly)	
Rogall (2021)	No interpretation	Perpetrators	
Cao and Chen (2022)	No	Rebellions per million population in 1600	They compute $\exp(\hat{\beta}) - 1$ and multiply by the baseline mean, then interpret this as the effect in levels

Appendix Table 1: Papers in the *AER* estimating effects for $\text{arcsinh}(Y)$ with selected quotes

Note: this table lists papers in the *AER* estimating treatment effects for $\text{arcsinh}(Y)$. The second column classifies papers by whether they interpret the units of the treatment effect as a percent/elasticity, with categories “yes”, “no”, or “no interpretation given.” The third column describes the units of the outcome before applying the arcsinh transformation, and the final column provides selected quotes and notes about the interpretation of the estimates. See [Section 2.3](#) for details.

Appendix Figure 1: t -statistics for effect on $\text{arcsinh}(Y)$, versus extensive margin t -statistic



Note: this table shows the t -statistic for the extensive margin effect on the x -axis, and the t -statistic for the treatment effect using $\text{arcsinh}(Y)$ on the y -axis. The circle shows the t -statistic using the original units, whereas the arrow shows the change if we first multiply the units by 100 before applying the arcsinh transformation. We omit two papers where there is no extensive margin. The plot shows that the t -statistics are close to the 45 degree line when the extensive margin is not close to zero, and tend to become closer when the units are made larger.

Paper	Treatment Effect Using:			Change from rescaling units:	
	$\log(1 + Y)$	$\log(1 + 100 \cdot Y)$	Ext. Margin	Raw	%
Azoulay et al (2019)	0.002	0.015	0.003	0.012	529
Fetzer et al (2021)	-0.138	-0.410	-0.059	-0.272	197
Johnson (2020)	-0.139	-0.408	-0.057	-0.269	194
Carranza et al (2022)	0.166	0.415	0.055	0.249	149
Cao and Chen (2022)	0.032	0.076	0.010	0.044	136
Rogall (2021)	1.109	2.015	0.195	0.906	82
Moretti (2021)	0.041	0.067	0.000	0.026	64
Berkouwer and Dean (2022)	-0.412	-0.484	0.010	-0.072	17
Arora et al (2021)	0.110	0.111	-0.001	0.001	1
Hjort and Poulsen (2019)	0.354	0.354	0.000	0.001	0

Appendix Table 2: Change in estimated treatment effects using $\log(1 + Y)$ from re-scaling the outcome by a factor of 100 in papers published in the *AER*

Note: this table repeats the exercise in Table 1 but replacing $\operatorname{arcsinh}(Y)$ with $\log(1 + Y)$ as the outcome in the second column, and $\operatorname{arcsinh}(100Y)$ with $\log(1 + 100Y)$ in the third column. The fourth column shows the estimated extensive margin effect, which is identical to the fourth column of Table 1. The final two columns show the raw difference and percentage difference between the second and third columns. The rows are sorted based on the percentage differences. Among the papers surveyed, which by construction report at least one specification using $\operatorname{arcsinh}(Y)$, Arora et al. (2021); Fetzer et al. (2021); Moretti (2021); Rogall (2021) also report specifications that contain $\log(1 + Y)$ on the left-hand side, and Johnson (2020) reports a specification with $\log(c + Y)$ on the left-hand side, where c is the first nonzero percentile of the distribution of the observed outcome variable.

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